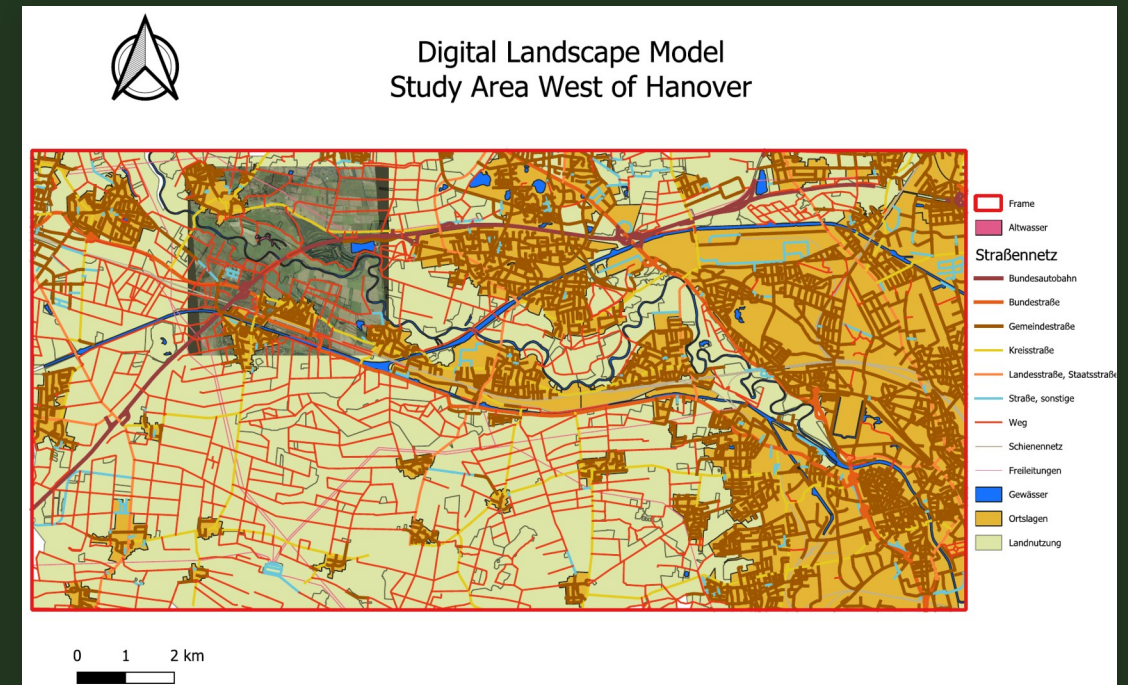


Wind Farm Suitability Analysis

West of Hanover

Data Preparation · Georeferencing · Digitization

Nadia Farsiani



The Digital Landscape Model (Map 1) — the final deliverable

The task, and my part in it

In an area of about **200 km² west of Hanover**, suitable areas for wind turbines are to be identified from a Digital Landscape Model (DLM). My stage prepares the data the rest of the analysis is built on.

1

Setup & CRS

New project; 7 layers; EPSG:31467

2

Georeference

Aerial photo → map coordinates

3

Digitize

Trace the Leine oxbow arms

4

DLM map

Assemble & export Map 1

1

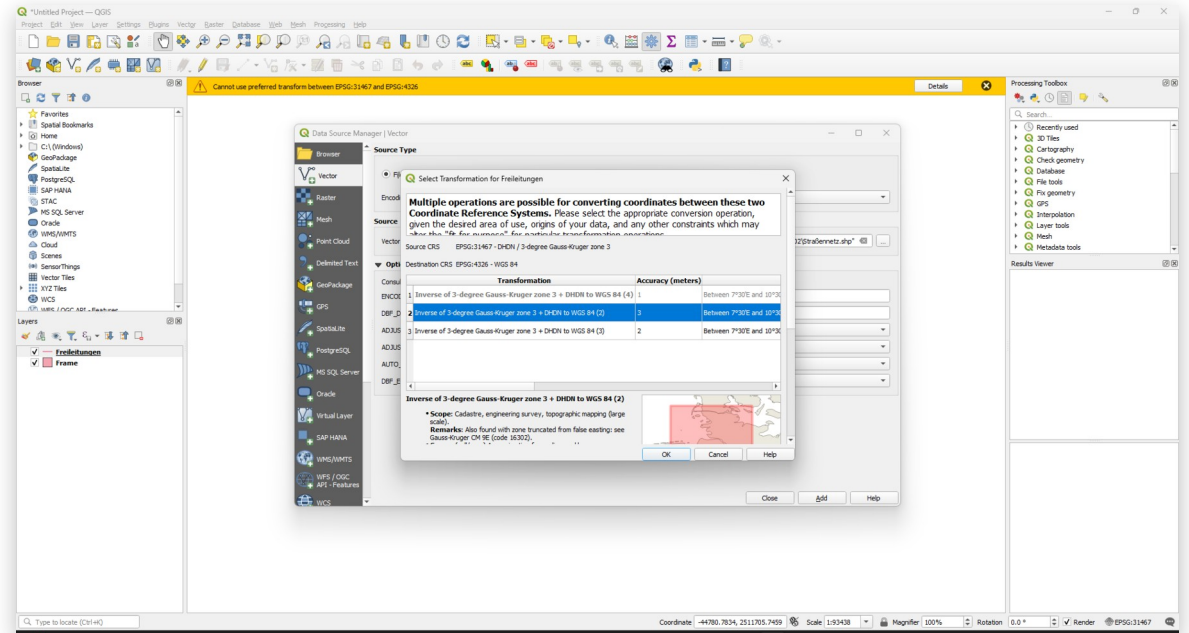
STEP 1 OF 5

Project setup and coordinate system

- Created a new QGIS project and loaded all seven shapefiles: study-area frame, roads, railways, power lines, water bodies, land use and places.
- Checked the coordinate system: every layer is in EPSG:31467 (Gauss-Krüger zone 3) — a projected system measured in metres.
- The project adopted this CRS from the data, so no reprojection was needed.
- All distances (15–650 m) and areas are therefore measured in true metres directly.

EPSG:31467

projected coordinate system · unit = metre

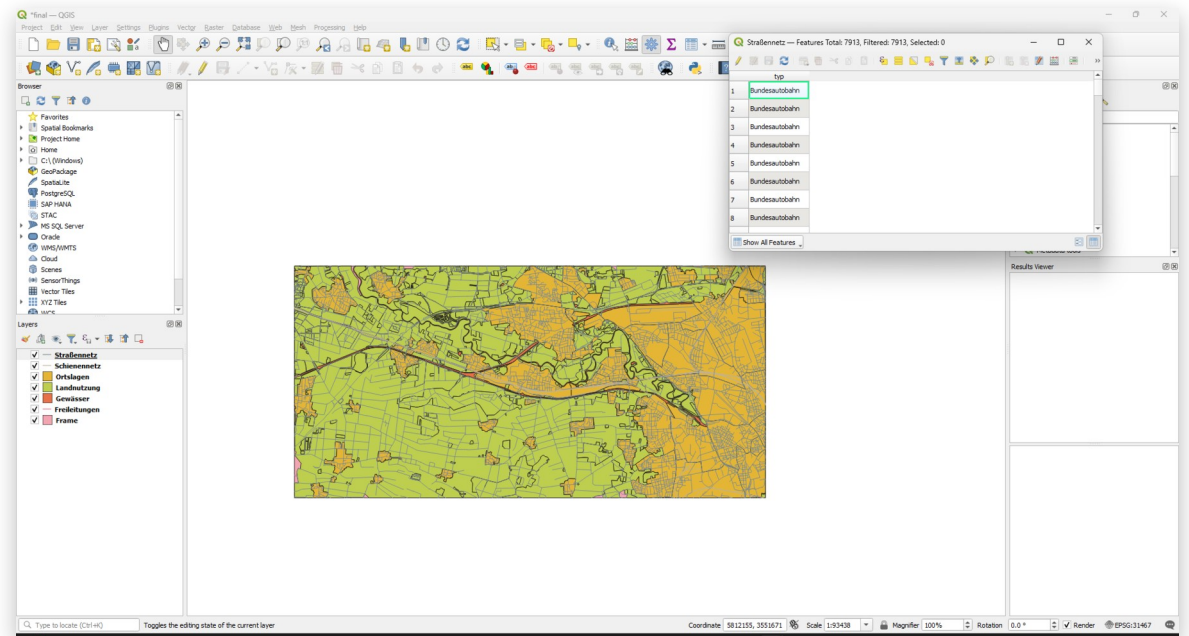


2

STEP 2 OF 5

Checking the data and its attributes

- Opened the attribute table of each layer to find the fields the later analysis depends on.
- Road class is stored in the field typ (Kreisstraße, Bundesstraße, Landesstraße/Staatsstraße, Bundesautobahn).
- Land use, water and railways are likewise classified in a typ field.
- Key finding: the settlement types are held in the land-use layer — Ortslagen contains only place names.



3

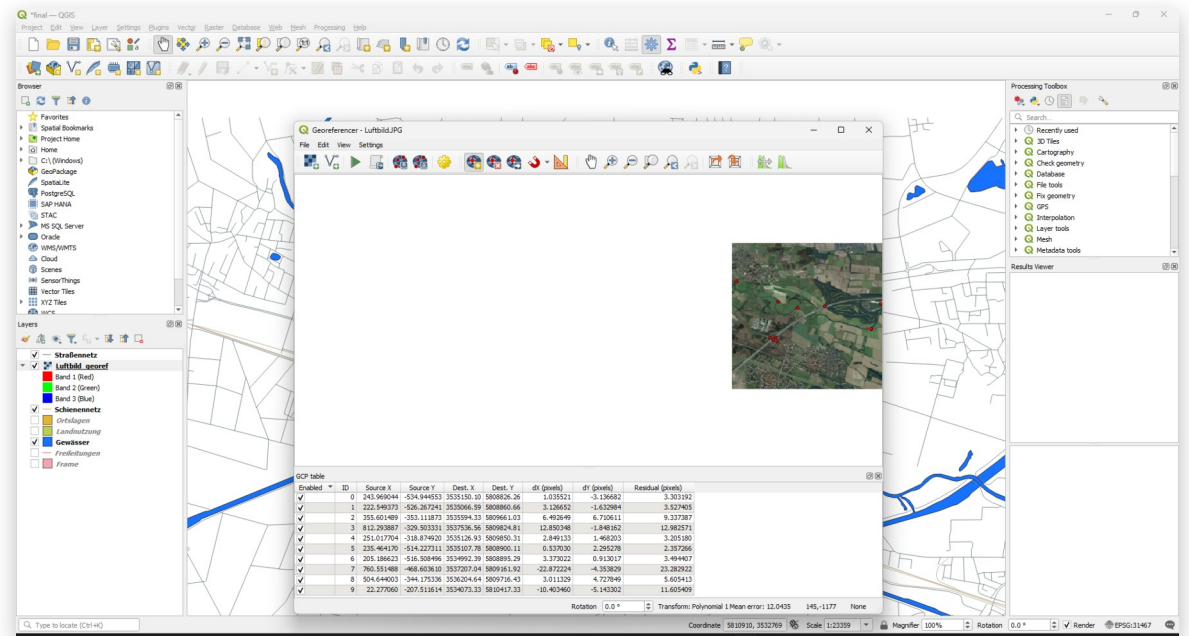
STEP 3 OF 5

Georeferencing the aerial photograph

- The supplied aerial photograph had no spatial reference.
- Placed ground control points by matching features in the photo (road junctions, water edges) to the vector layers.
- Transformation: Polynomial 1 (affine), nearest-neighbour resampling, target EPSG:31467.
- 10 points placed; 1 outlier disabled, leaving 9 used for the fit.

5.1 px

mean error $\approx$ 20–25 m on the ground



4

STEP 4 OF 5

Digitizing the oxbow arms of the Leine

The cut-off meander loops of the Leine are not in the water layer, so I traced them as new polygons. Result: **7 polygons in Altwasser.gpkg** (field typ = 'Altwasser').



Aerial photograph — the oxbow arms are visible



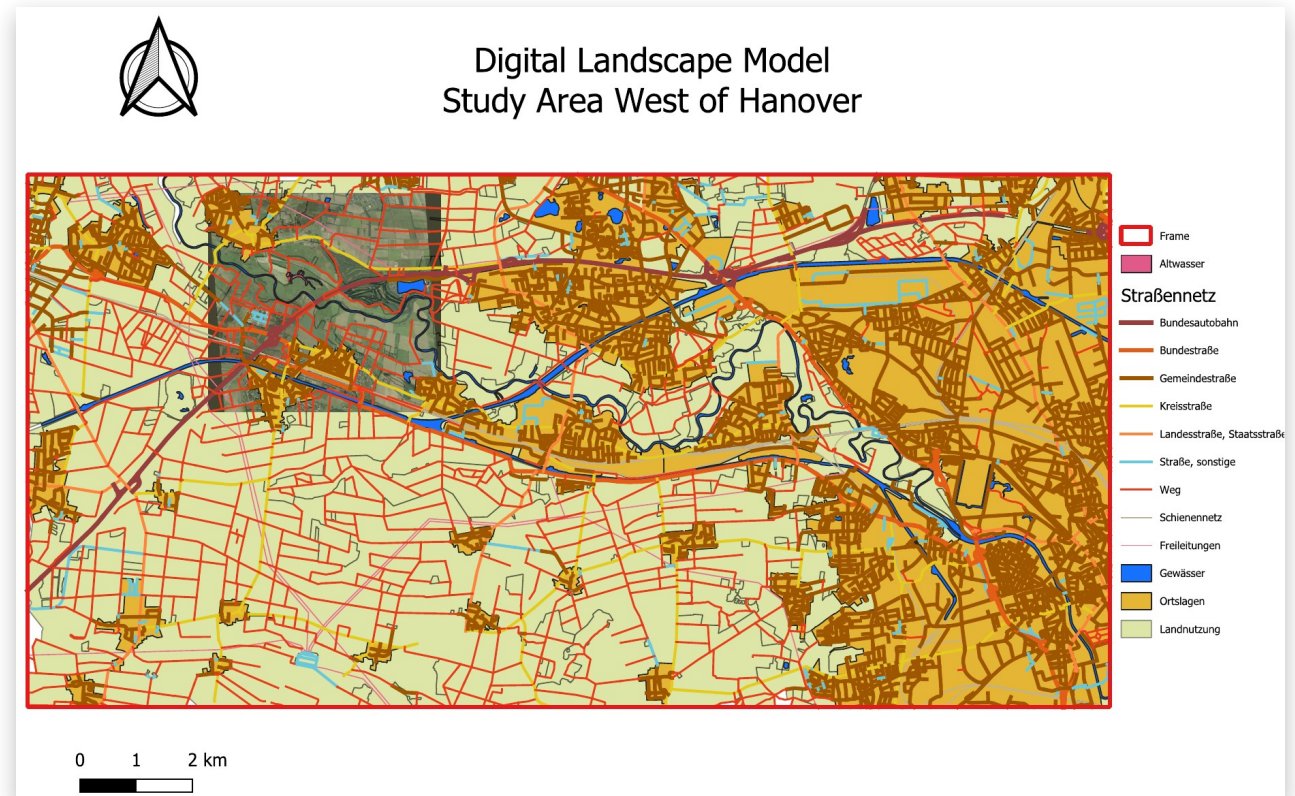
Digitized result — 7 polygons over the aerial

5

STEP 5 OF 5

Producing the Digital Landscape Model map

- Assembled all data layers, the georeferenced aerial and the digitized oxbows in the QGIS print layout.
- Styled the road network by class (the typ field).
- Added the four required map elements: title, legend, scale bar and north arrow.
- Exported the finished layout as DLM.pdf.



Results & outputs

7

layers prepared
in EPSG:31467

5.1 px

georeferencing
mean error

7

oxbow polygons
digitized

Map 1

DLM map
produced

Files produced

final.qgz

the complete QGIS project

All files in one folder · relative paths · coordinate system EPSG:31467 throughout

Luftbild_georef.tif

the georeferenced aerial photograph

Luftbild.points

the georeferencing control points

Altwasser.gpkg

the 7 digitized oxbow polygons

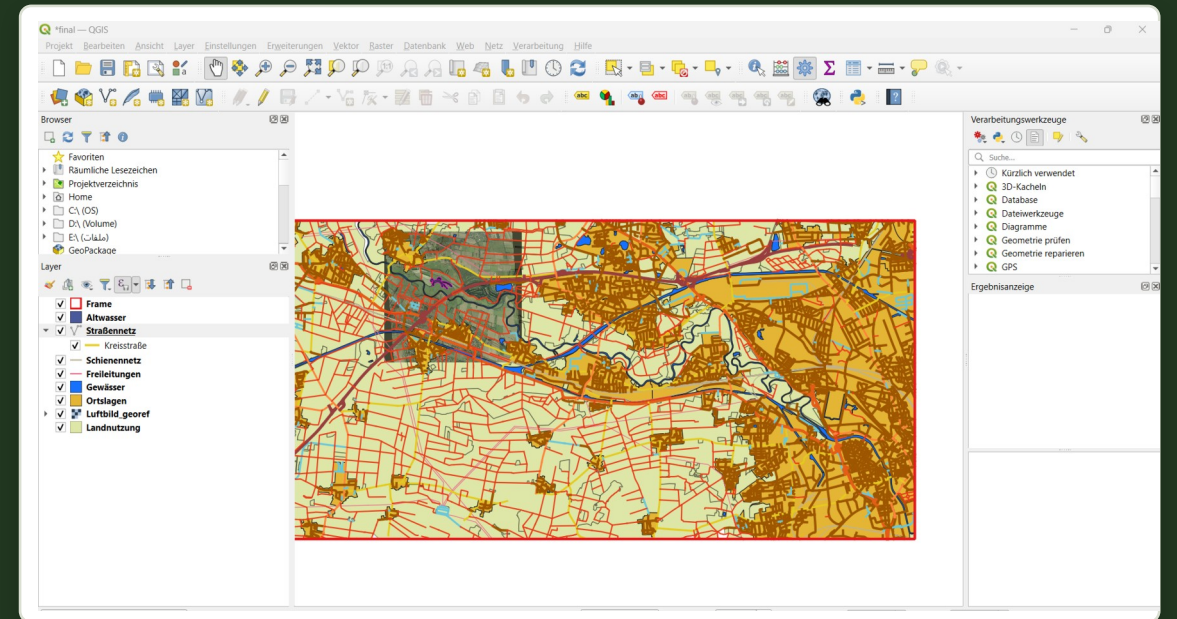
DLM.pdf

the exported Digital Landscape Model map

Buffer Analysis and Exclusion Areas

West of Hanover

Safety Buffers · Settlement Classification · Total Exclusion Area



The task, and my part in it

Person 1 handed over the prepared DLM: seven layers, the georeferenced aerial photo, and the digitized oxbow arms. My stage turns every distance rule in the assignment into an actual exclusion layer — the mask Person 3 will subtract farmland from.

1

Check the data

one field, typ — and two compound values

2

Build 11 buffers

11 distance rules → 11 layers

3

Split settlements

spatial join solves a missing attribute

4

Combine

Merge → Fix → Dissolve → Clip

5

Check & hand over

Total_Exclusion_Area → Person 3



All eleven safety buffers, merged and colour-coded by class

96.4%

of the study area ends up excluded

11 → 1

buffers dissolved into one clean mask

1

STEP 1 OF 5

Checking the data first

- Every class I need — road type, land use, water, rail — lives in a single field: `typ`. Nothing is split across separate attributes.
- `Landnutzung.shp` holds 2,919 polygons across 20 unique `typ` values. `Ortslagen.shp`, by contrast, carries only place names — no settlement-type field at all.
- Key finding: some values are compound strings. The semicolon in 'Wald; Forst' and the comma in 'Landesstraße, Staatsstraße' are part of the value itself, not a separator between two classes.
- That means every selection has to match the string verbatim — no LIKE wildcards, no splitting on punctuation, or the filter silently returns nothing.

typ

the one field every selection is built on

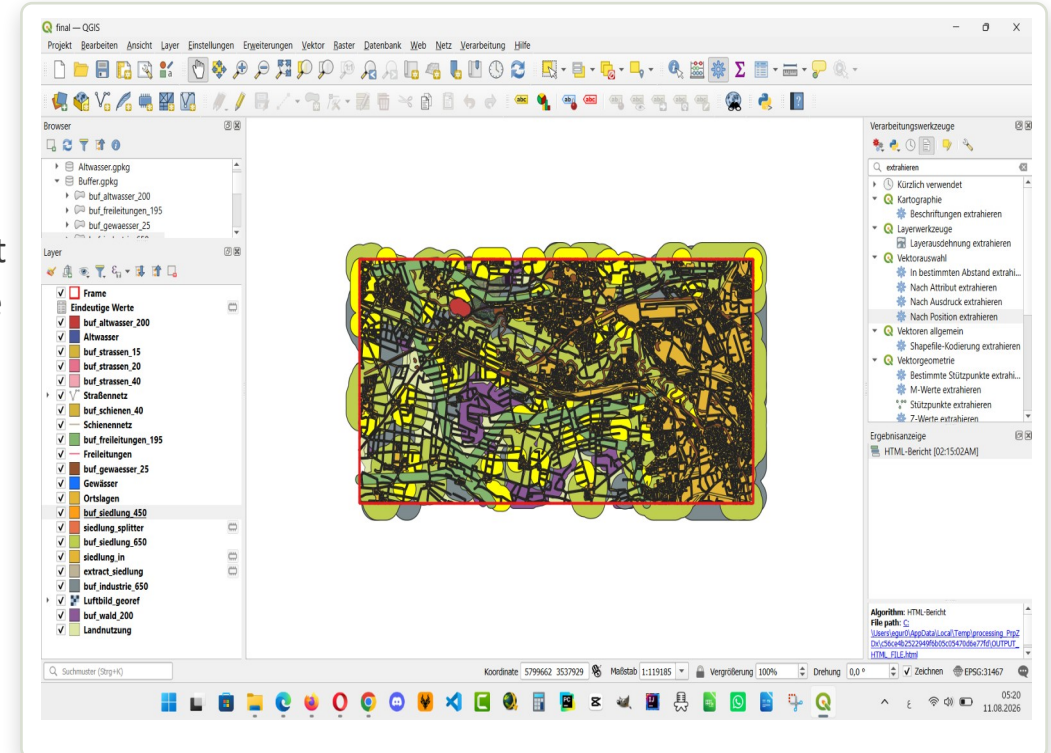
Select strings verbatim — no LIKE, no splitting.

1

STEP 1 OF 4

Building the eleven safety buffers

- Straßennetz.shp keeps every road class in one field (typ) — I only buffered the three categories the assignment names (Kreis-, Bundes-/Landes-, Bundesautobahn) and left Gemeindestraße and Weg untouched.
- Distances range from 15 m for district roads up to 650 m for settlements — set with the Buffer tool, "dissolve result" switched on so same-class buffers merge automatically.
- Same settings on every run for consistency: 16 segments per quarter-circle, round caps and joins.
- Forest and oxbow arms (protected biotopes) both get a 200 m buffer; overhead power lines get 195 m — three times the 65 m rotor diameter.



All 11 buffers before merging

11 buffers

one per constraint class, 15 m to 650 m

1

STEP 1 OF 4

Building the eleven safety buffers

Feature	Buffer	Selection used
District roads (Kreisstraße)	15 m	typ = 'Kreisstraße'
Federal / state roads	20 m	typ IN ('Bundesstraße', 'Landesstraße, Staatsstraße')
Motorways (Bundesautobahn)	40 m	typ = 'Bundesautobahn'
Railway lines	40 m	all features
Overhead power lines	195 m	all features (3 × 65 m rotor diameter)
Water bodies	25 m	all features
Oxbow arms (Altwasser)	200 m	all features (protected biotope)
Forest	200 m	typ = 'Wald; Forst'
Settlement / industrial / commercial / special-function areas	650 m	Siedlungsflaeche intersecting Ortslagen, or typ IN ('Indus.-Gew.-Fl.', 'Fl. fkt. Praeg.')
Individual houses / settlement splinters	450 m	Siedlungsflaeche NOT intersecting Ortslagen



All 11 buffers merged, before Total_Exclusion_Area is built

The settlement classification problem

- The assignment splits settlements into two classes with different distances: settlements 'assigned to a local location' (650 m) and individual houses or splinters that are not (450 m).
- Landnutzung.shp doesn't carry that distinction — every settlement patch is just typ = 'Siedlungsflaeche', with nothing to tell the two apart.
- Solved it spatially instead of by attribute: any Siedlungsflaeche polygon touching a named village in Ortslagen counts as assigned (650 m); anything that doesn't touch one is a splinter (450 m).
- Result: 1,640 of 1,771 settlement polygons (≈3,231 ha) sit inside a named village; 131 (≈118 ha) stand on their own.

1,640 / 131

settlement patches: assigned vs. splinter

650 / 450 m

buffer distance for each class

No attribute solved this — a spatial join with Ortslagen did.

3

STEP 3 OF 4

Merging everything into one exclusion layer

1

Merge

all 11 buffer layers combined into a single layer

2

Fix geometries

cleaned up before dissolving, not after — repairing first keeps the topology clean going into the dissolve

3

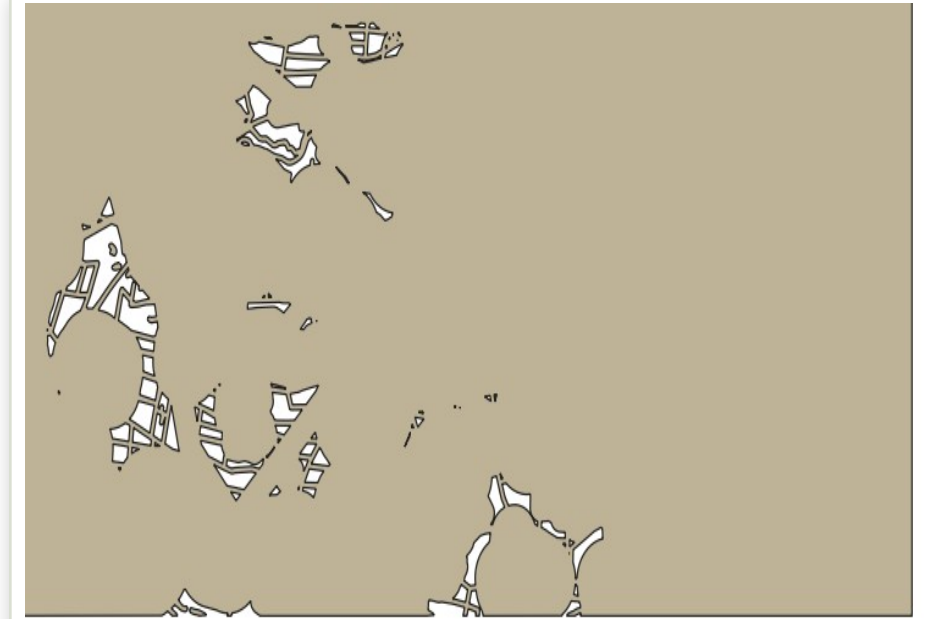
Dissolve

no grouping field, so every touching buffer melts into one shape with no leftover internal lines

4

Clip

cut to the Frame boundary, so buffers sticking out past the study area don't count



Total_Exclusion_Area

One feature- a single MultiPolygon object

Results & outputs

Being precise here mattered: the assignment's own '~200 km²' is a rough figure. The Frame boundary actually measures 181.78 km² — and that's the number the final percentage is built on.

181.78 km²

study area (Frame),
measured, not estimated

96.4%

of the area excluded
once every buffer applies

658 ha

left over — scattered
candidate patches for Person 3

1

final exclusion feature,
clipped to the Frame

Files produced

Total_Exclusion_Area.gpkg — final exclusion layer — main deliverable

Buffer.gpkg — all 11 buffers + merged/dissolved intermediates, kept for traceability

Altwasser.gpkg / Luftbild_georef.tif — carried over from Person 1

Key expressions & formulas

Copy-ready, and verified against the data before use

1 • Selection expressions

<code>"typ" = 'Kreisstraße'</code>	→ 15 m
<code>"typ" IN ('Bundesstraße', 'Landesstraße, Staatsstraße')</code>	→ 20 m
<code>"typ" = 'Bundesautobahn'</code>	→ 40 m
<code>"typ" = 'Wald; Forst'</code>	→ 200 m
<code>"typ" IN ('Indus.-Gew.-Fl.', 'Fl. fkt. Praeg.')</code>	→ 650 m
<code>"typ" = 'Siedlungsflaeche'</code>	→ extract

2 • Spatial split (Extract by location)

<code>Siedlungsflaeche n Ortslagen</code>	(intersects)	→ 650 m
<code>Siedlungsflaeche \ Ortslagen</code>	(disjoint)	→ 450 m

3 • Area & exclusion rate

$A_{ha} = \$area / 10000$

$P_{ex} = (17,519.48 / 18,177.8) \times 100 \approx 96.4\%$

4 • Tested, then removed — no effect

`$area < 10,000,000`

- largest forest polygon: 261.8 ha
- largest settlement polygon: 22.0 ha
- both far below the threshold, so the filter excluded nothing

Verbatim strings only

no LIKE, no wildcards, no splitting on ';' or ','